

P5: Straight Lines and Circles

1. Straight Lines

Gradient

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Equation Forms

Form	Equation	When to use
Slope-intercept	$y = mx + c$	When gradient and y-intercept are known
Point-slope	$y - y_1 = m(x - x_1)$	When gradient and one point are known
General form	$ax + by + c = 0$	When integers are preferred; used in circle questions

Parallel and Perpendicular Lines

Relationship	Condition
Parallel	$m_1 = m_2$ (equal gradients)
Perpendicular	$m_1 \times m_2 = -1$, i.e., $m_2 = -\frac{1}{m_1}$

Example: Find the equation of the line perpendicular to $y = 3x + 1$ that passes through $(6, 2)$.

Gradient of given line: $m_1 = 3$

Perpendicular gradient: $m_2 = -\frac{1}{3}$

Equation: $y - 2 = -\frac{1}{3}(x - 6) \Rightarrow y = -\frac{1}{3}x + 4$

Midpoint and Distance

Midpoint of (x_1, y_1) and (x_2, y_2) : $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$

Distance: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

2. Circles

Equation of a Circle

$$(x - a)^2 + (y - b)^2 = r^2$$

Centre (a, b) , radius r .

General form: $x^2 + y^2 + 2gx + 2fy + c = 0$

Centre: $(-g, -f)$; Radius: $r = \sqrt{g^2 + f^2 - c}$

(Complete the square to convert to standard form)

Example: Find the centre and radius of $x^2 + y^2 - 6x + 4y - 3 = 0$.

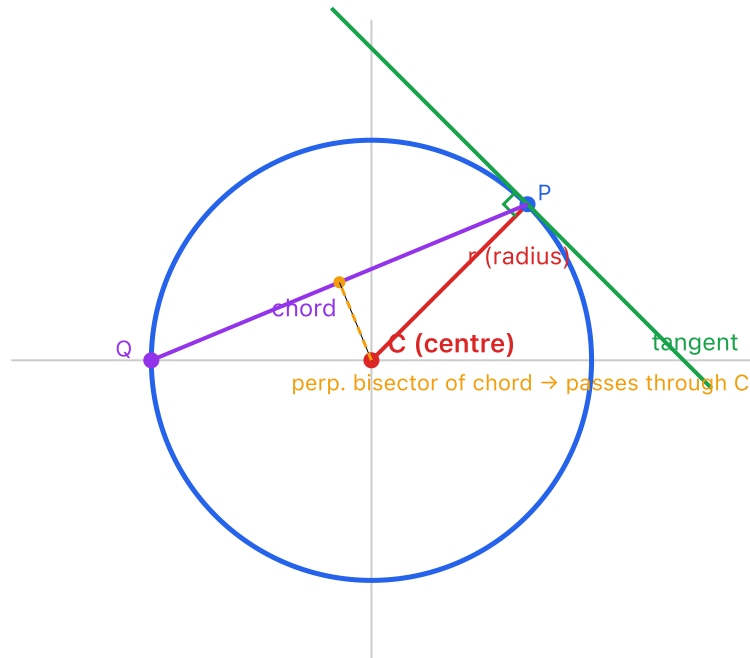
Complete the square:

$$(x^2 - 6x + 9) + (y^2 + 4y + 4) = 3 + 9 + 4$$

$$(x - 3)^2 + (y + 2)^2 = 16$$

Centre $(3, -2)$, radius 4.

Circle Diagram



Tangent to a Circle

The tangent at point P on a circle is perpendicular to the radius CP .

Example: Find the equation of the tangent to the circle $(x - 3)^2 + (y + 2)^2 = 25$ at the point $(7, 1)$.

Centre: $(3, -2)$. Gradient of radius to $(7, 1)$: $m_r = \frac{1 - (-2)}{7 - 3} = \frac{3}{4}$

Gradient of tangent: $m_t = -\frac{4}{3}$

Equation: $y - 1 = -\frac{4}{3}(x - 7) \Rightarrow 3y - 3 = -4x + 28 \Rightarrow 4x + 3y = 31$

Chord Bisector

The perpendicular bisector of any chord of a circle passes through the centre. This property is frequently examined.

Example: Show that the circle $(x - 1)^2 + (y - 3)^2 = 50$ passes through $A(6, 8)$ and $B(-4, 8)$, and find the equation of the chord AB.

Check A: $(6 - 1)^2 + (8 - 3)^2 = 25 + 25 = 50 \checkmark$

Check B: $(-4 - 1)^2 + (8 - 3)^2 = 25 + 25 = 50 \checkmark$

Both points have $y = 8$, so the chord AB is $y = 8$. Its perpendicular bisector is $x = 1$, which passes through the centre $(1, 3) \checkmark$

3. Circle Geometry Theorems

Theorem	Statement
Tangent perpendicular to radius	A tangent at point P is perpendicular to the radius at P
Angle in a semicircle	Any angle in a semicircle (angle subtended by a diameter at the circumference) is 90°
Perpendicular bisector of chord	The perpendicular bisector of any chord passes through the centre
Equal tangents from external point	Two tangents drawn from an external point to a circle are equal in length

Example (Angle in semicircle): Points $A(-2, 0)$ and $B(6, 0)$ are the ends of a diameter. Show that $\angle APB = 90^\circ$ for the point $P(2, 4)$ on the circle.

Check: if AB is a diameter, then any point P on the circle satisfies $\angle APB = 90^\circ$.

Circle centre $(2, 0)$, radius $|AB|/2 = 4$. If $P(2, 4)$: $\overrightarrow{PA} = (-4, -4)$, $\overrightarrow{PB} = (4, -4)$. Dot product: $(-4)(4) + (-4)(-4) = -16 + 16 = 0$. $\checkmark$ Right angle confirmed.

Exam Tip: If a question says "AB is a diameter", immediately note that any point C on the circle gives $\angle ACB = 90^\circ$. This is the key to many circle geometry proofs.

4. Summary

Formula	Use
$m = \frac{y_2 - y_1}{x_2 - x_1}$	Gradient between two points
$y - y_1 = m(x - x_1)$	Line through point with gradient
$m_1 m_2 = -1$	Perpendicular lines
$(x - a)^2 + (y - b)^2 = r^2$	Circle, centre (a, b) , radius r
Complete the square	Convert general circle to standard form
Tangent $\perp$ radius	Find tangent gradient at a point